



University of Amsterdam
Recommender Systems

Data, bias and automation

Dr. Dina Strikovic | Meeting 2



Contents

- Recap: new tech and datafication
 - What's the RS connection?
- Automation of daily lives
 - What are the implications?
- Data and automation bias





Can you solve this riddle?





Remember “New Tech and Datafication”?

Recap

- Our lives have become about not just about *how* we go through our days, but also how our lives become data
- New sets of actors, infrastructures and (unintended) consequences in a data-rich digital society
- Ethical principles must guide the assessment of large and small decisions *before, during* and *after* their implementation
- Data is not neutral – who's in charge of the decisions (about what to collect, how to collect it, etc) is important!
 - Incentives → outcomes.



From data to algorithm

Algorithms as a key feature of our society

1. Patterns of inclusion

The choices behind what makes it into an index in the first place, what is excluded, and how data is made algorithm ready

2. Cycles of anticipation

The implications of algorithm providers' attempts to thoroughly know and predict their users, and how the conclusions they draw can matter

3. The evaluation of relevance

The criteria by which algorithms determine what is relevant, how those criteria are obscured from us

Algorithms as a key feature of our society

4. The promise of algorithmic objectivity

The way the technical character of the algorithm is positioned as an assurance of impartiality, and how that claim is maintained in the face of controversy

5. Entanglement with practice

How users reshape their practices to suit the algorithms they depend on

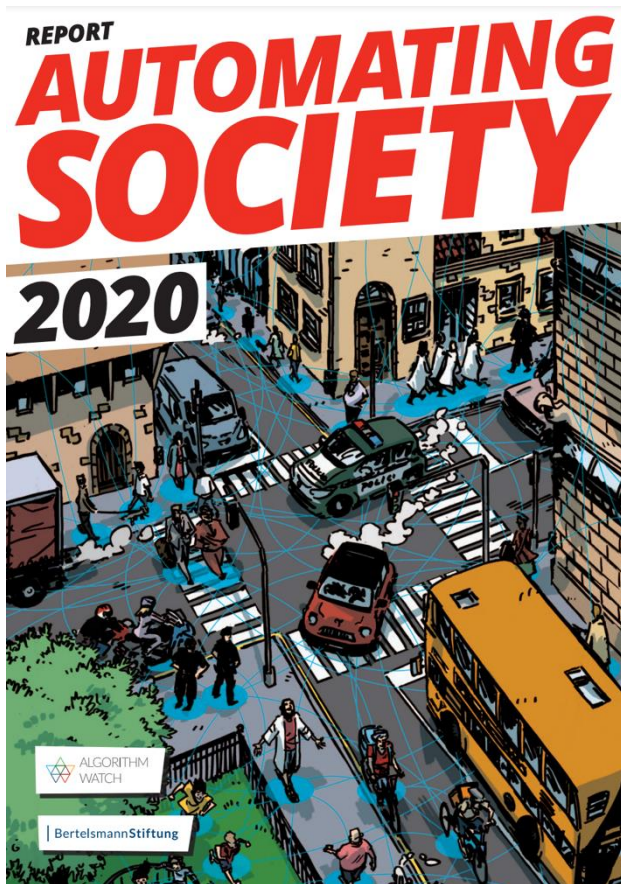
6. The production of calculated publics

The algorithmic presentation of publics back to themselves shapes a public's sense of itself, and who is best positioned to benefit from that knowledge



From datafication → Automation of our lives

Automation in our daily lives



(AlgorithmWatch, 2020)

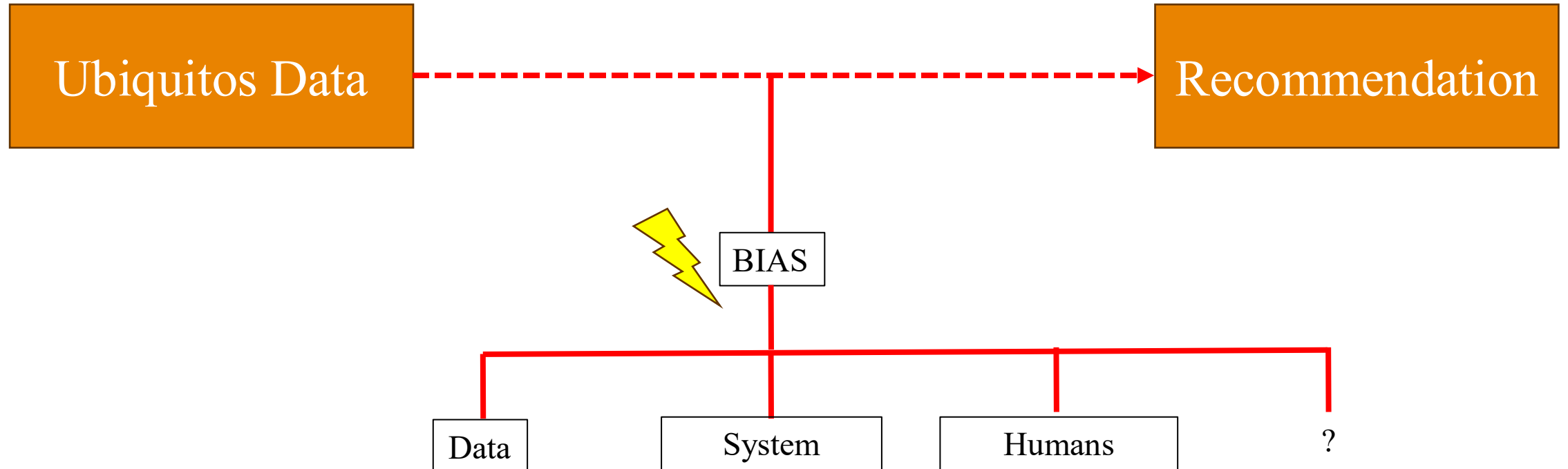
Automation already present in several aspects
our lives – even if we do not see it.

Examples in the Netherlands:

- Court and judiciary settings
- Usage by municipal governments
- Health care
- Financial trading
- Law enforcement

Automation in our daily lives

- Trust the process?

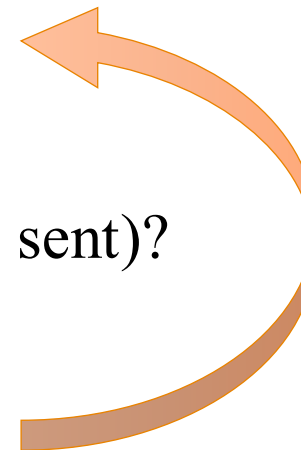
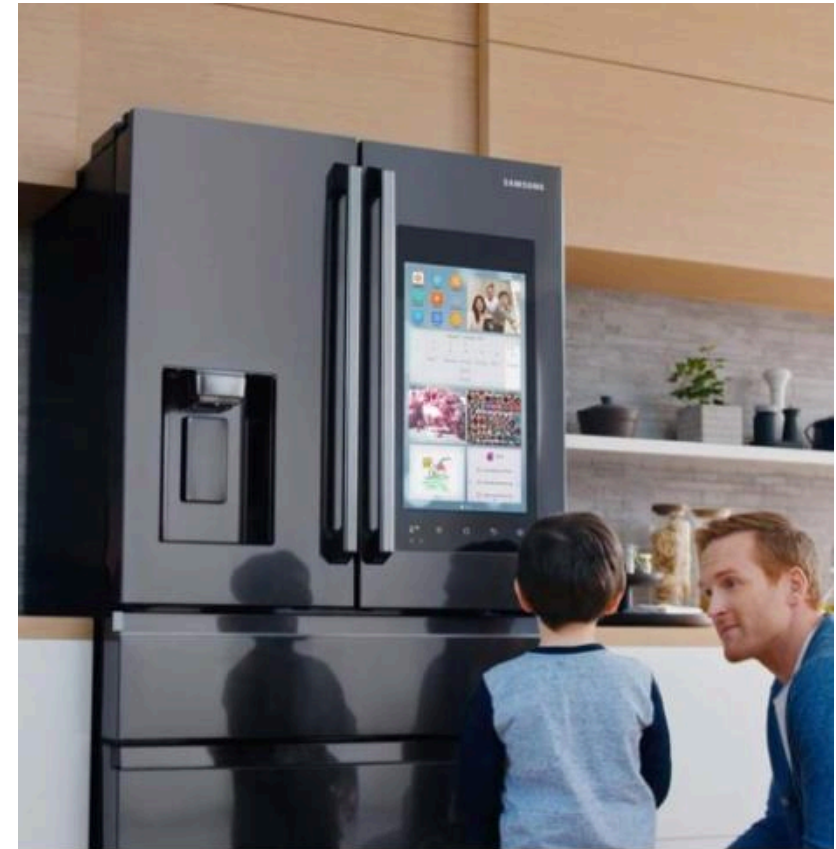




Data Bias

Data bias: data is not neutral

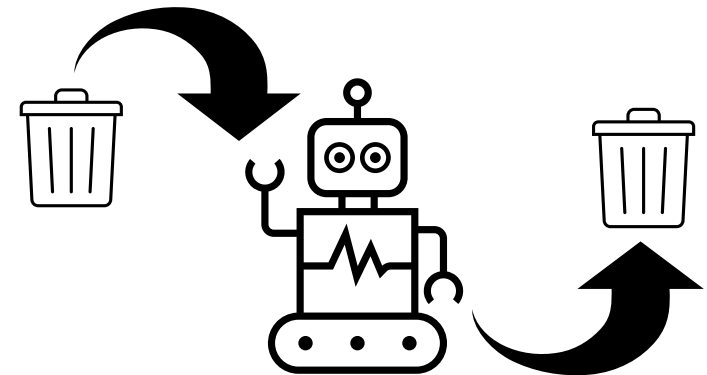
- Incentives from those who collect the data
- There is always a “pay-off” behind it
- Urges us to ask the questions of...
 - Where (is the data being collected)?
 - Who (is collecting the data)?
 - When (is the data actively/passively being sent)?
 - Why is this important to the collector?
 - How can the data be turned into value?

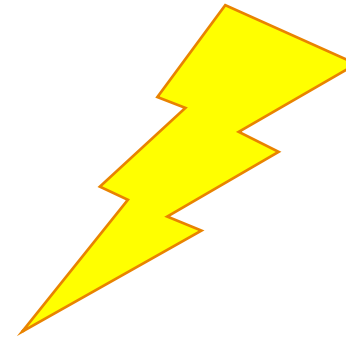
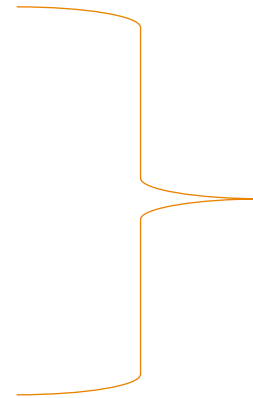
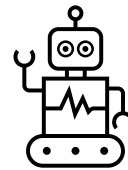
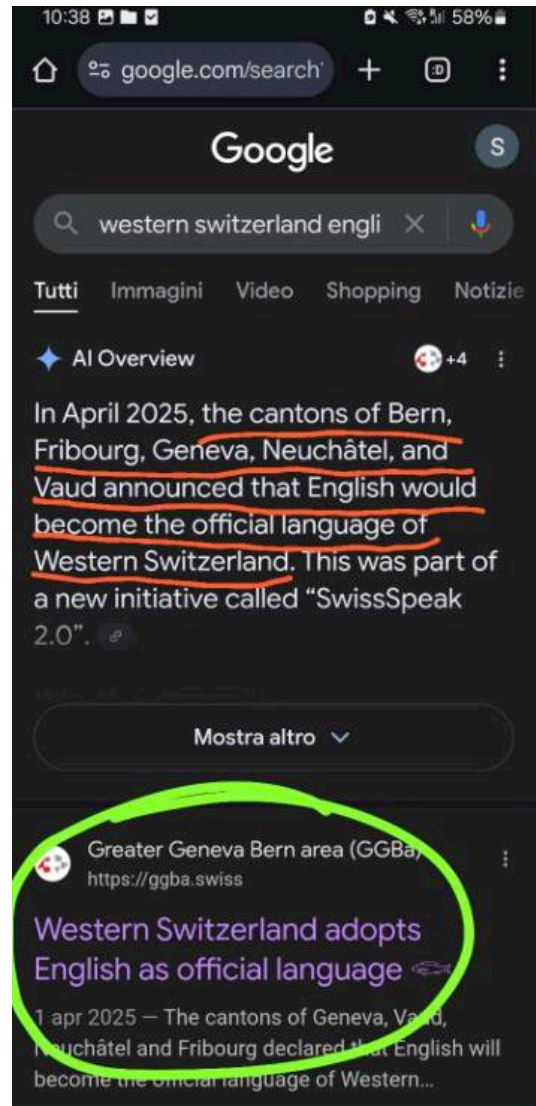


Data bias: data is not ~~neutral~~ fair

- How can data be biased/unfair?
- Certain groups generate more data than others
- Inequalities encoded within the data
- Some groups are underrepresented

“Garbage in, garbage out.”







Letter | Published: 25 January 2017

Dermatologist-level classification of skin cancer with deep neural networks

[Andre Esteva](#) , [Brett Kuperl](#) , [Roberto A. Novoa](#) , [Justin Ko](#), [Susan M. Swetter](#), [Helen M. Blau](#) & [Sebastian Thrun](#) 

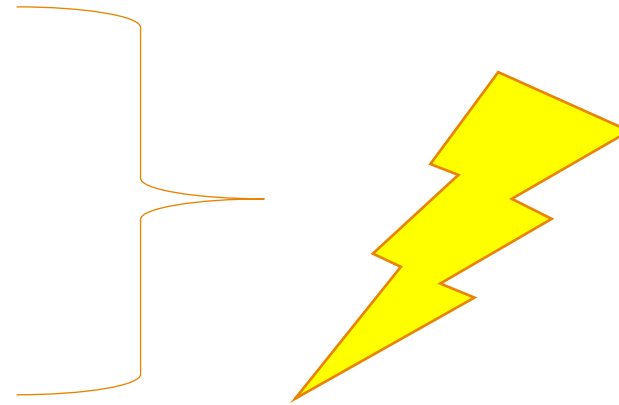
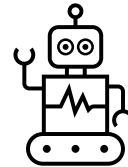
[Nature](#) **542**, 115–118 (2017) | [Cite this article](#)

244k Accesses | **8973** Citations | **2858** Altmetric | [Metrics](#)

 A [Corrigendum](#) to this article was published on 29 June 2017

Abstract

Skin cancer, the most common human malignancy^{1,2,3}, is primarily diagnosed visually, beginning with an initial clinical screening and followed potentially by dermoscopic analysis, a biopsy and histopathological examination. Automated classification of skin lesions using images is a challenging task owing to the fine-grained variability in the appearance of skin





COMPAS

JAMES RIVELLI

Prior Offenses

1 domestic violence
aggravated assault, 1
grand theft, 1 petty
theft, 1 drug trafficking

Subsequent Offenses

1 grand theft

ROBERT CANNON

Prior Offense

1 petty theft

Subsequent Offenses

None

VERNON PRATER

Prior Offenses

2 armed robberies, 1
attempted armed
robbery

Subsequent Offenses

1 grand theft

BRISHA BORDEN

Prior Offenses

4 juvenile
misdemeanors

Subsequent Offenses

None



COMPAS

JAMES RIVELLI	ROBERT CANNON
Prior Offenses 1 domestic violence aggravated assault, 1 grand theft, 1 petty theft, 1 drug trafficking	Prior Offense 1 petty theft
Subsequent Offenses 1 grand theft	Subsequent Offenses None

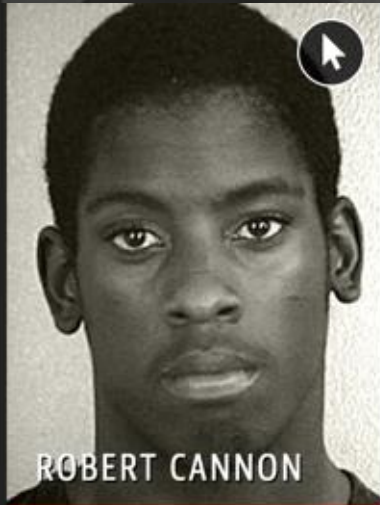
VERNON PRATER	BRISHA BORDEN
Prior Offenses 2 armed robberies, 1 attempted armed robbery	Prior Offenses 4 juvenile misdemeanors
Subsequent Offenses 1 grand theft	Subsequent Offenses None

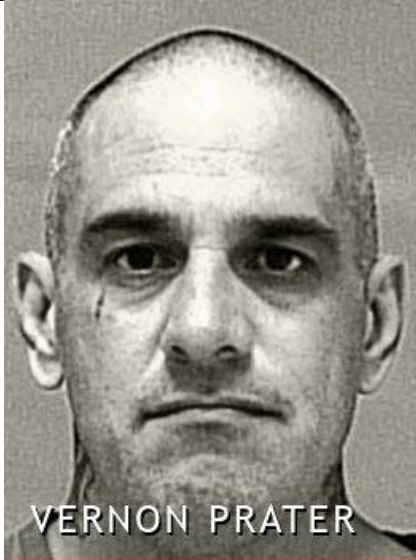
LOW RISK 3 | MEDIUM RISK 6

LOW RISK 3 | HIGH RISK 8



COMPAS

JAMES RIVELLI Prior Offenses 1 domestic violence aggravated assault, 1 grand theft, 1 petty theft, 1 drug trafficking Subsequent Offenses 1 grand theft	ROBERT CANNON Prior Offense 1 petty theft Subsequent Offenses None
 JAMES RIVELLI LOW RISK 3	 ROBERT CANNON MEDIUM RISK 6

VERNON PRATER Prior Offenses 2 armed robberies, 1 attempted armed robbery Subsequent Offenses 1 grand theft	BRISHA BORDEN Prior Offenses 4 juvenile misdemeanors Subsequent Offenses None
 VERNON PRATER LOW RISK 3	 BRISHA BORDEN HIGH RISK 8

Data bias: data is not ~~neutral~~ fair

Just remove/restrict access to sensitive data! (?)

- COMPAS did not have access to information about race
- Data bias can occur even when we are “careful”
- Algorithms will pick up on this via *proxies*...

Data bias: data is not ~~neutral~~ fair

Just remove/restrict access to sensitive data! (?)

Accessible data...	... can be a proxy for...
Postal code	Race, income, immigration status
Educational	Income, age, socio-economic status
Criminal history	Race (systemic bias), income
Health conditions	Age, gender
Family size	Religion, income
Extracurricular activities	Race, economic status, gender, education



Data bias: data is not ~~neutral~~ fair

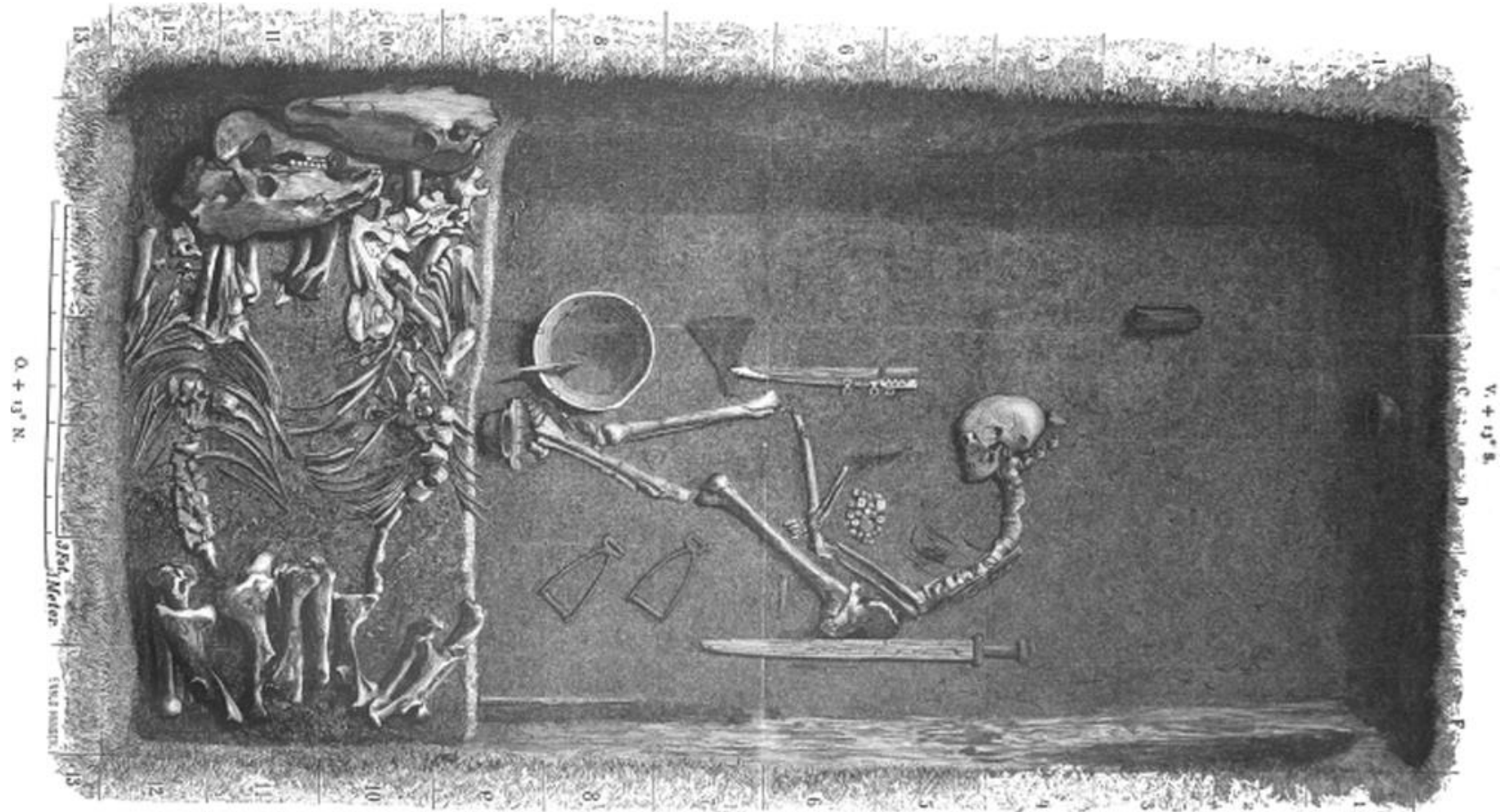
Just remove/restrict access to sensitive data! (?)

- Correlated variables remain!
- Machine learning models are good at re-constructing hidden variables...
- ... that can carry with them systemic bias.

Variable	Proxies in your data?
Gender	
Age	
Ethnicity	
Socio-economic status	
Occupation	

What does data bias look like?

Gender bias



What does data bias look like?

Gender bias



What does data bias look like?

Gender bias





What does data bias look like?

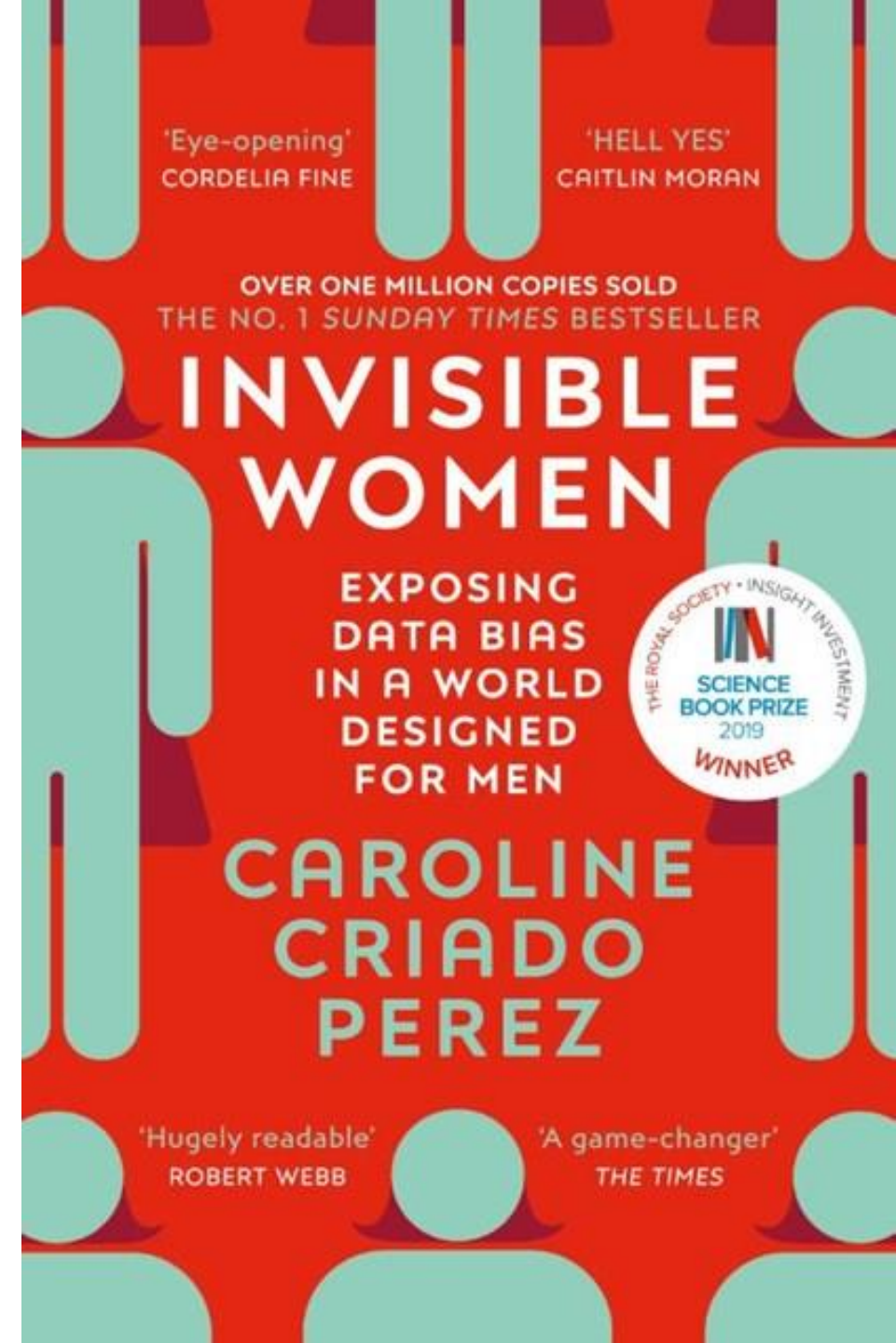
Gender bias





What does data bias look like?

Gender bias



What does data bias look like?

Bias in the (training) data reinforces biases

- Amazon software for recruitment → famous (?) scandal
- Specifically asked to not look at gender and race
- What do you think happened?

What does data bias look like?

Bias in the (training) data reinforces biases

- Amazon software for recruitment
- Specifically asked to not look at gender and race
- What do you think happened?



HELLO
my name is

Jared*

*therefore, I'm rich!

What does data bias look like?

What about other bias?

IN GROUPS: Think of another kind of data bias that could influence algorithms

- Take some time to researcht this. Where can we find data gaps?
- Where does the bias occur? At what decision point?
- What implications could this have on the system that is trained on it?
- What are the possible outcomes??

→ Present this to the class 

What does data bias look like?

In terms of recommenders

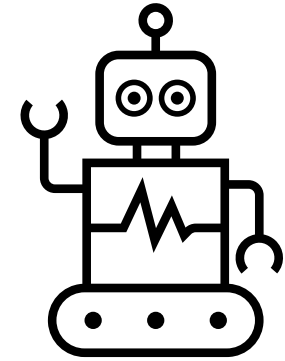
- Mostly men clicked on this kind of job → show vacancies to men.
- Educational data on immigrants → show ads for vocational schools.
- Lower-income people click on short-term credit offers → show them more.
- Higher CTR for low-educated people on clickbait → offer sensational news.
- Postal code associated with incidents → offer higher insurance rates.



System Bias

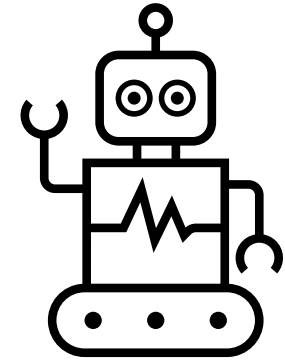
Bias in the system

- Biases can also occur on the systems level
- Even with “neutral” and “fair” data, design choices matter



Bias in the system

- Biases can also occur on the systems level
 - Even with “neutral” and “fair” data, design choices matter
 - Decisions about the programming of the recommender
 - Optimization and learning
 - Ranking objectives
 - Etc....
- Recommenders seldom optimize for fairness – they optimize for measurable business goals.



Bias in the system

- **Popularity bias:** systems tend to recommend already popular items, which marginalizes niche and less-exposed content.
- **Exposure bias:** what the system shows affects what users interact with and thus what gets learned (by system) — creating feedback loops.
- **Stereotyping and miscalibration:** more complex systems can produce recommendations that reflect stereotypical associations (e.g., by gender or age), disproportionately affecting minority groups.

Bias in the system

Netflix (for example)

- 1000 movies = 850 (Hollywood) + 150 (minor studios)
- Minority studio production have a much lower CTR
- BUT, users mostly click on recommended items

Is this model
biased?



Metric	Model A	Model B
CTR recommended items	12%	
Minority Exposure	5%	
Watch share	4%	





Bias in the system

Netflix (for example)

- 1000 movies = 850 (Hollywood) + 150 (minor studios)
- Minority studio production have a much lower CTR
- BUT, users mostly click on recommended items
 - exposure $\neq$ database
 - visibility = power

Which model is
better long-
term?

Metric	Model A	Model B
CTR recommended items	12%	10.5%
Minority Exposure	5%	30%
Watch share	4%	22%



Human Bias



Can you solve this riddle?

Maggie and Maureen are born to the same mother and father, on the same day, of the same month of the same year. They are not twins.

Human vs. Machine (bias)

What's worse?

- A) Hiring manager favors candidates from his own university?
- B) Algorithm ranks candidates lower because historical hires from certain universities didn't perform well?

- Pro-publica audited COMPAS
 - Much harder to do with human bias



Bias propagated by humans

Even with neutral data + good systems

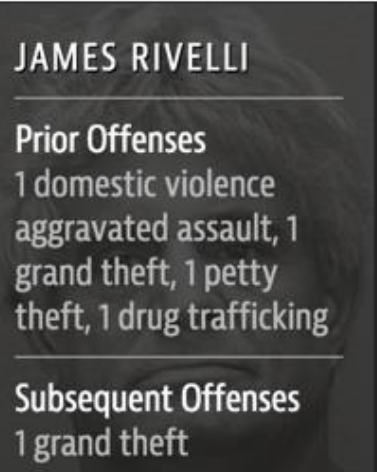
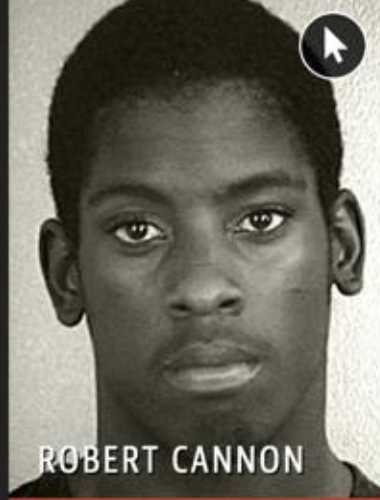
- Human decision making in the end → good thing?

How these systems are utilized in institutions becomes an important question.



Bias propagated by humans

There is still human agency (bias?) in the process

 JAMES RIVELLI Prior Offenses 1 domestic violence aggravated assault, 1 grand theft, 1 petty theft, 1 drug trafficking Subsequent Offenses 1 grand theft	 ROBERT CANNON Prior Offense 1 petty theft Subsequent Offenses None
 JAMES RIVELLI LOW RISK 3	 ROBERT CANNON MEDIUM RISK 6

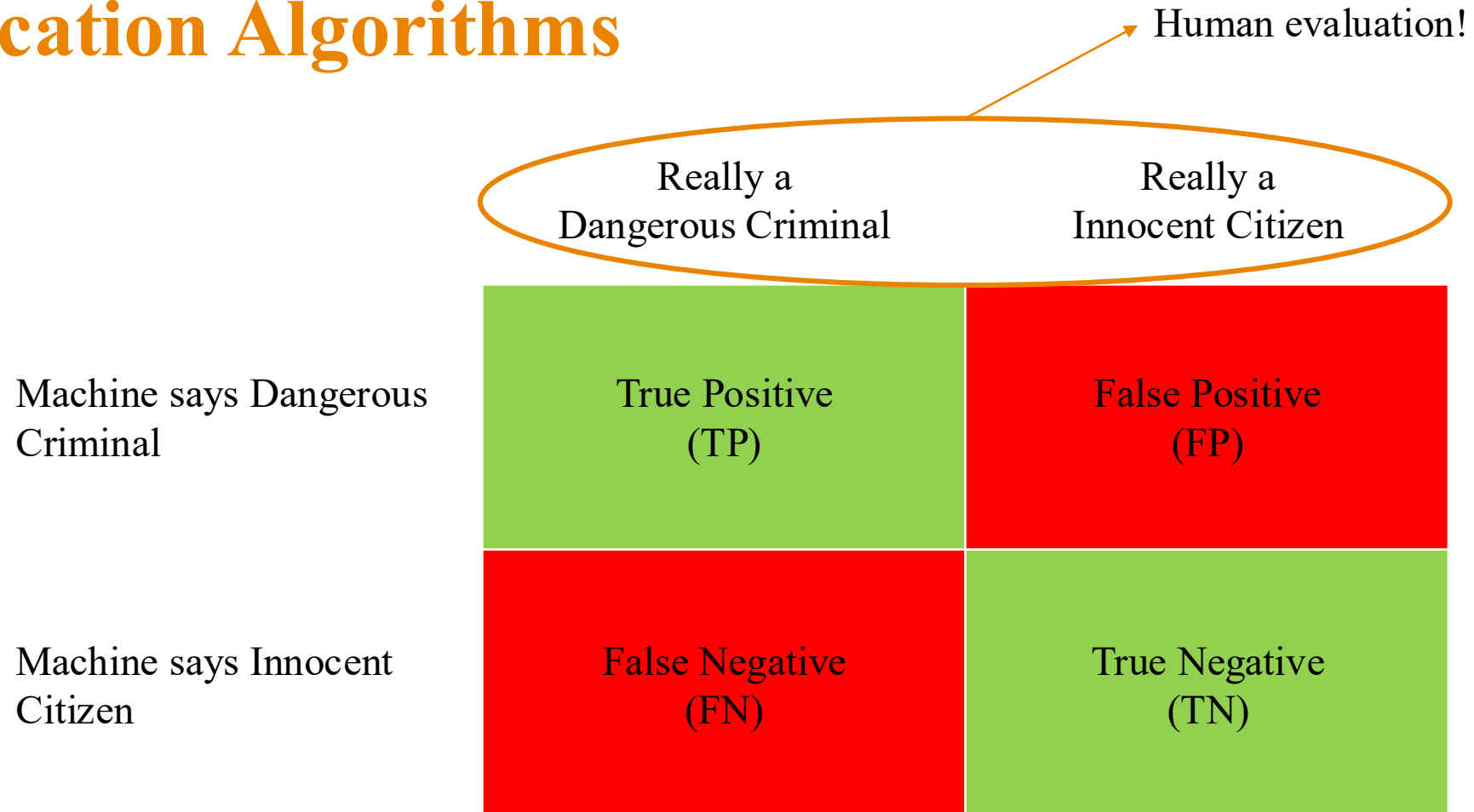


Classification Algorithms

	Really a Dangerous Criminal	Really a Innocent Citizen
Machine says Dangerous Criminal	True Positive (TP)	False Positive (FP)
Machine says Innocent Citizen	False Negative (FN)	True Negative (TN)

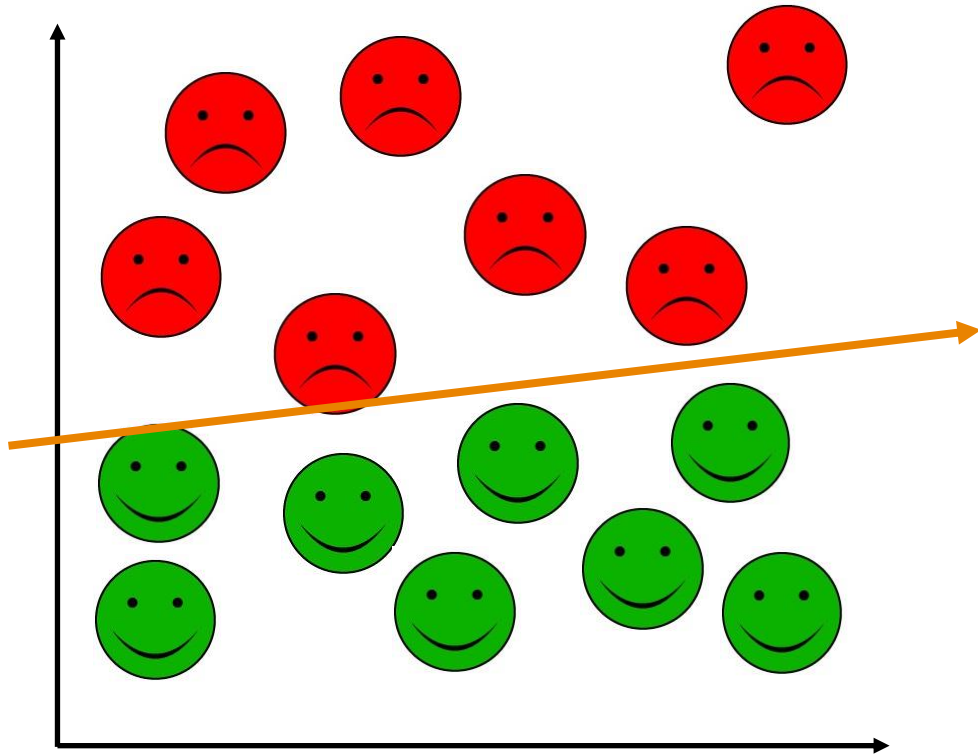


Classification Algorithms



Classification Algorithms

Variable Y



Variable X



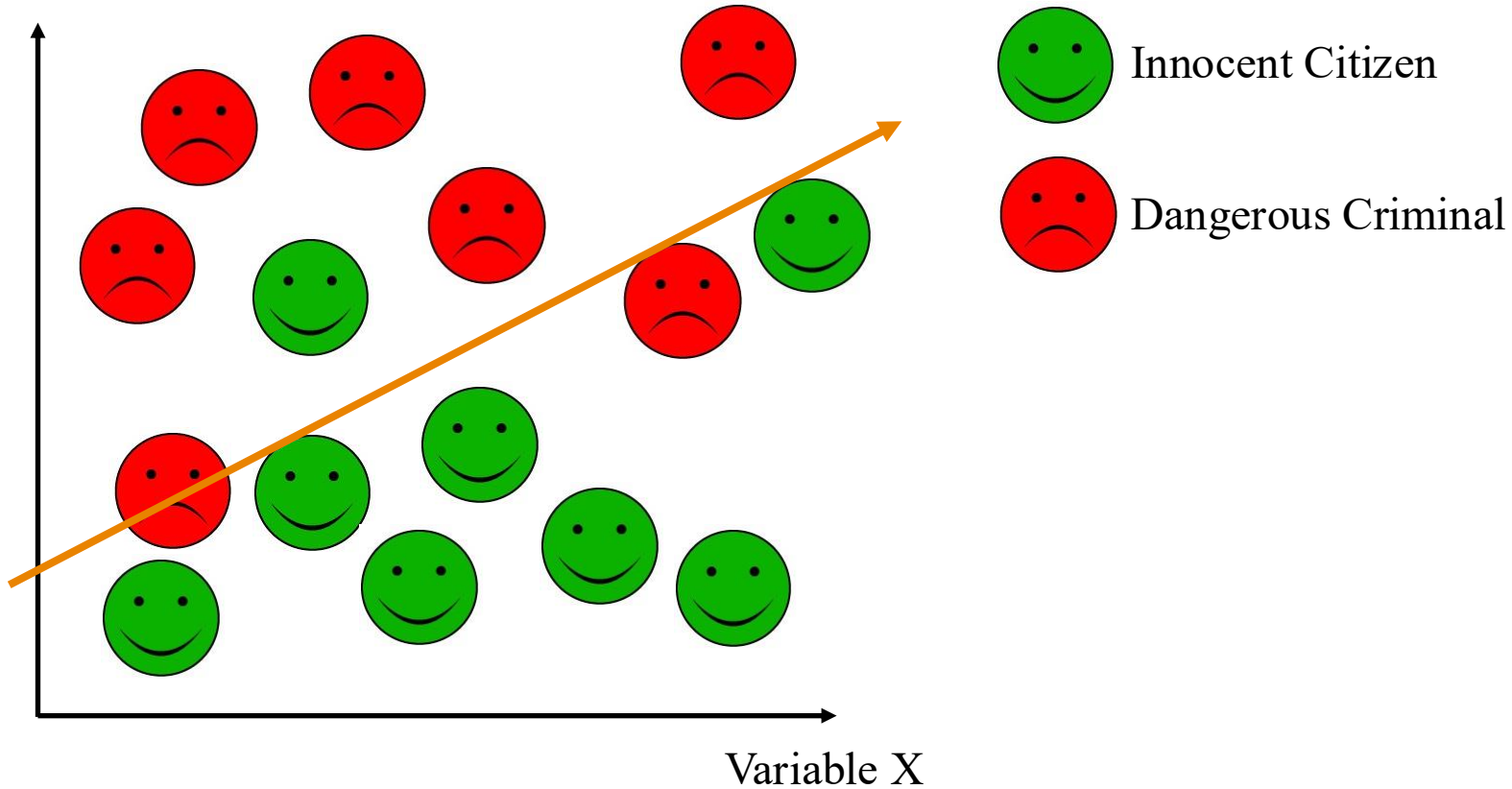
Innocent Citizen



Dangerous Criminal

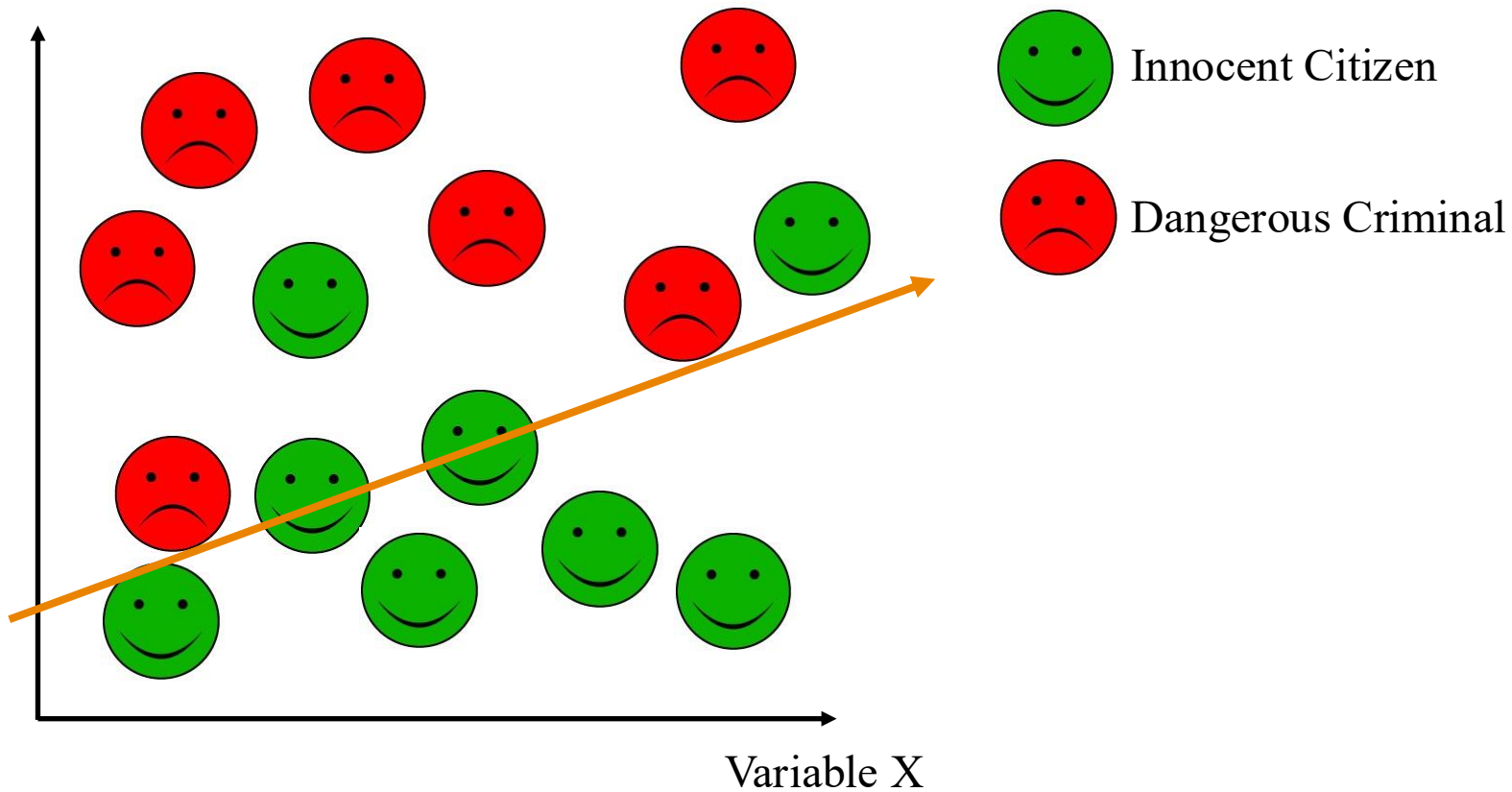
Classification Algorithms

Variable Y



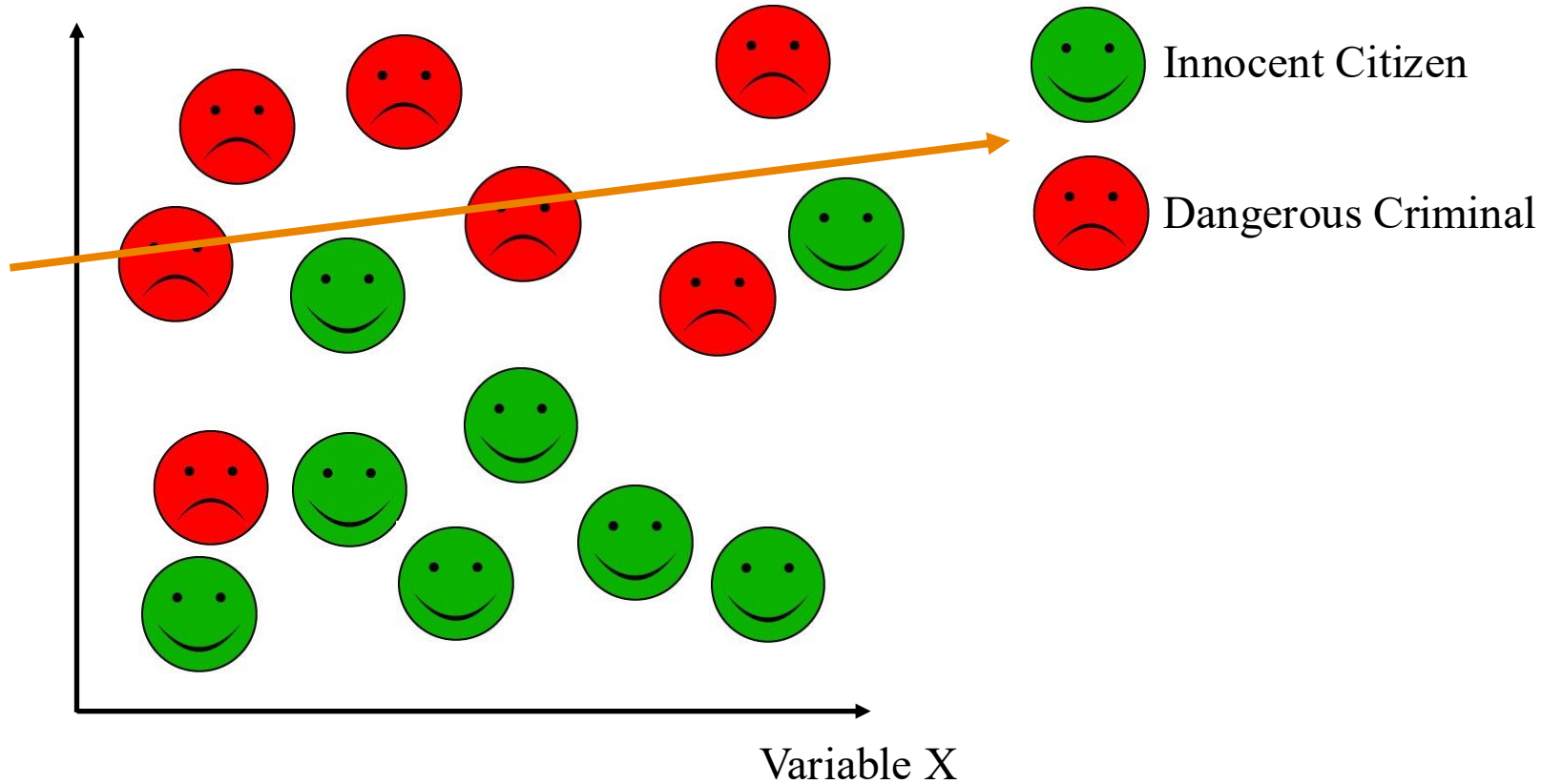
Classification Algorithms

Variable Y



Classification Algorithms

Variable Y

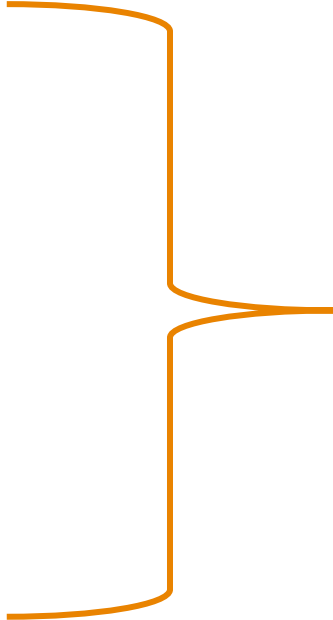


Human vs. Machine (bias)

Trade-offs with the ways systems are designed

- Tweaks in data/system/human agency → concessions:

- Accuracy
- Fairness
- Explainability
- Balance
- Bias
- Safety
- Privacy



*What are the better
compromises to make?*



Summary

- Our lives have become automated in many more ways than we may be aware of
- New sets of data and biases are influencing those automation processes: data is not de facto fair.
- Considerations about implementation and optimization of algorithms have to consider data biases
- Human-machine interactions play a role, too: no biases exist in isolation
 - And no biases are created without data that points to them